Mechanical Engineering Portfolio

Sera Arimori

Candidate for Bachelor of Science in Mechanical Engineering

Northeastern University

Contents

1. About Me
2. Previous Experience
    - Tesla Drive Unit Inverter
    - Silicon Synapse Lab
3. Projects
    - Vehicle Dynamics Modeling
    - Waddle Walk
    - Race to the Summit
    - Mini Golf Course
4. Contact Me
5. Links

About Me

Hi, I'm Sera, a third-year mechanical engineering student at Northeastern University.

I'm originally from Phoenix, Arizona, often visit family in Tokyo, Japan, and now go to school in Boston, Massachusetts, so I'm used to traveling often! I love animals, and I love to read and run.

In my career, I'm passionate about increasing sustainability in mechanical design, and I'm looking to utilize my previous experience at Tesla and in an aerospace robotics lab in a fast-paced environment that fosters my skills in design and hands-on manufacturing.

Please contact me at Arimori.s@northeastern.edu if you'd like to connect, ask questions, or simply are curious about any of my experiences!



Previous Experience



Tesla Drive Unit Inverter: Mechanical Design Co-op

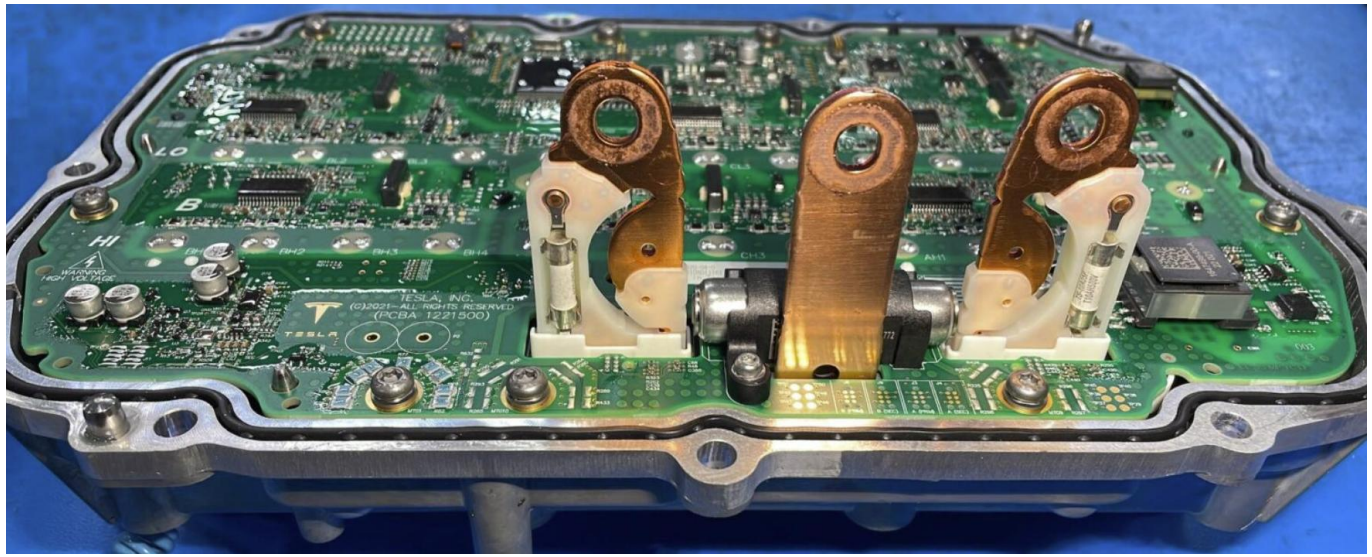
I worked in Palo Alto for 6 months, and I was able to own my project's entire timeline, which went from receiving a problem statement to presenting tangible results. Seeing the project from beginning to end taught me how to design solutions with manufacturability and testability in mind, which has made my iterative design process smoother and more robust.

Silicon Synapse Lab: Research Assistant

I worked on COBRA, a snake-like robot designed to collect ice deposits from the Moon. My work ranged from making design changes to previous prototypes to increase the TRL of the robot to making proposals to win funding.

Tesla: Drive Unit Inverter

Mechanical Design Co-op



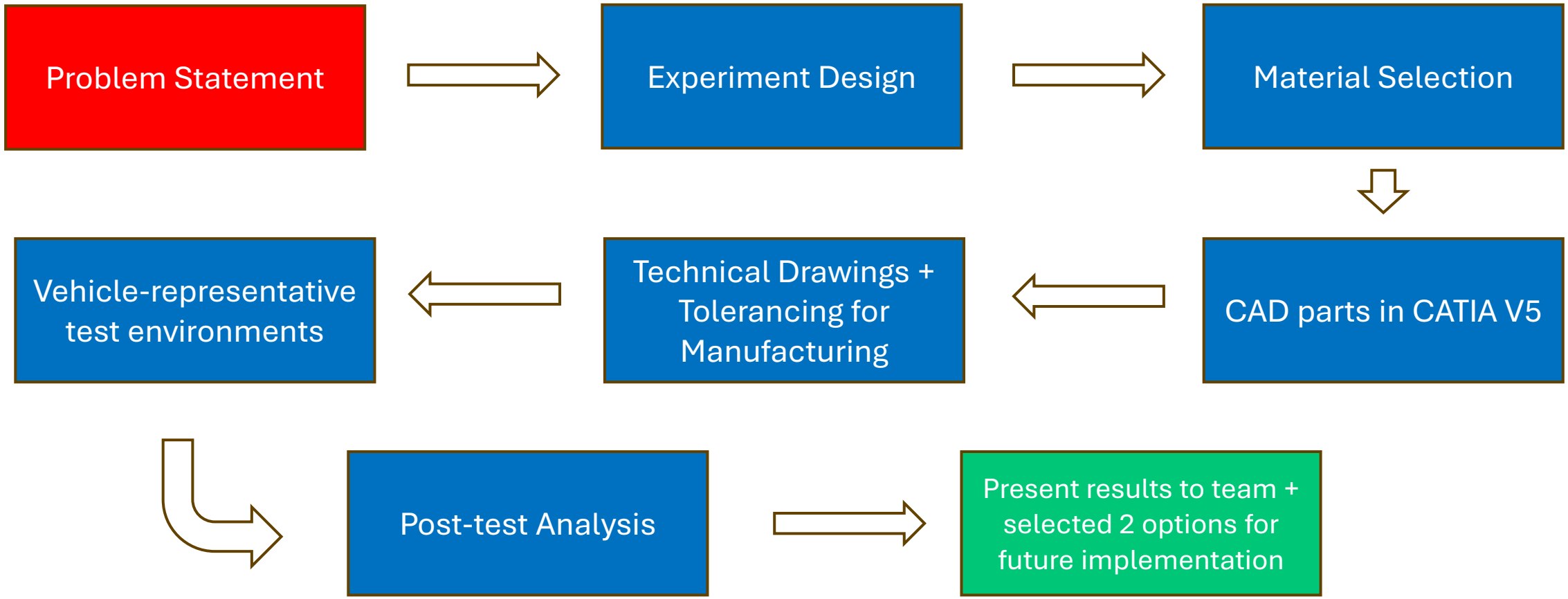
A drive inverter!
Going in, it was a completely foreign part to me, and I'm grateful to have been on such a fun and complex part.

Photo Credit: Drive Inverter - Rear (Remove and Replace)

Upon starting my internship at Tesla, I was immediately thrown into a fast-paced learning environment where I was given my project on my second day.

From then on, I was given free reign to design an experiment that derisks a key design decision; I was in charge of everything from designing the parts and interfacing with suppliers to make material selections to designing a test plan that achieves life-cycle testing in proper vehicle-representative environments. Post-test analysis was also a key learning opportunity in my internship; I was able to learn how to quantify pressure-decay, material degradation, and surface profiles to quantify my test results for best final presentation. Ultimately, by creating a pass/fail rubric, I down-selected from 5 materials to 2, identifying them as strong candidates for future design implementation.

Tesla: A Broad Overview of my Internship



Tesla: Skills

CAD:

- ***CATIA V5***: I used CATIA V5 for all of my parts along with some exposure to 3DX.

Prototyping/Experimentation:

- ***3D Printing***: I needed a lot of parts with a unique geometry quickly, so 3D printing seemed to be the best option.
- ***Material Application***: During my time, I got to troubleshoot a lot of failures, which meant I got to test a bunch of materials! It was fun seeing how many things can be tried; during this time, I learned how to select and apply thermal epoxies and heat shrinks.
- ***Python/DAQ Setup***: Some experiments had data best collected during the tests; for these, I set up instrumentation and a DAQ and wrote a Python script to collect data.

Post Testing Analysis:

- ***Degradation***: Post-test, I wanted to see how seals, materials, and adhesives degraded, meaning I learned how to perform pressure-decay tests, Hipot tests, and shear tests.
- ***Surfaces***: I learned how to use a profilometer, and when I realized I wanted the machine to account for unevenness of the part, a white-light interferometer to quantify surface roughness of my parts.

COBRA: Crater Observing Bio-Inspired Rolling Articulator



COBRA, a snake-like robot designed to collect ice deposits from the Moon's Shackleton Crater, was originally designed for NASA's 2022 Breakthrough, Innovative and Game-changing Idea Challenge, and after winning 1st place, the lab continues to develop the robot.

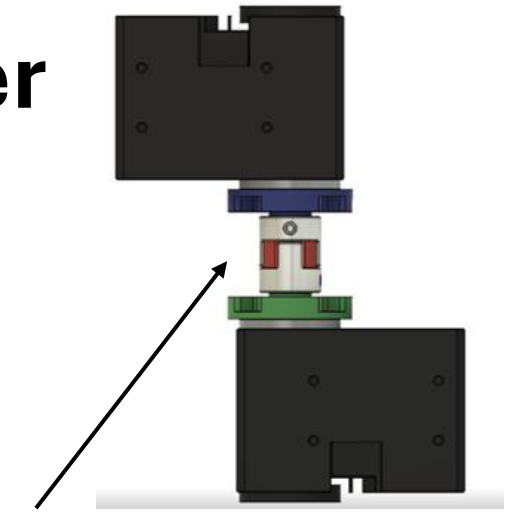
I joined while the lab was looking to advance the first 3D-printed PLA prototype into a metal version.

I learned an incredible amount, beginning with research into what design considerations are made for the extreme lunar environment to designing and editing body modules to enhance the robot's performance.

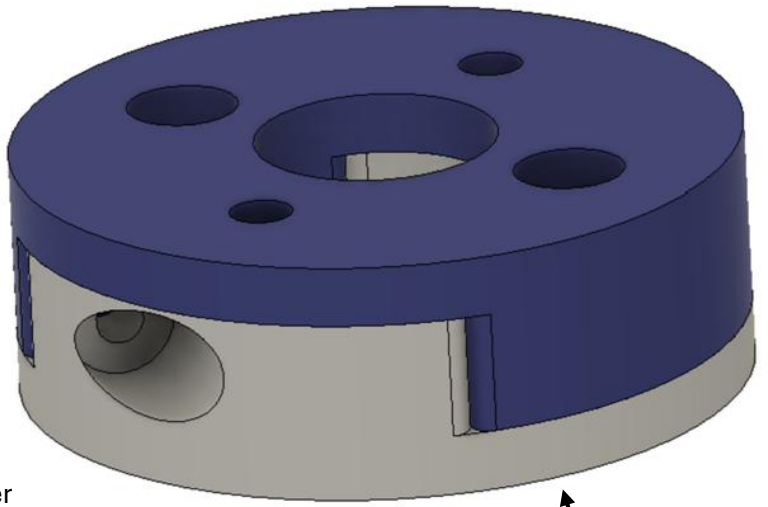
Photo Credit: Award-Winning Robot Could Help NASA Find Water on the Moon

COBRA Joint Motor Coupler

- **Problem**: COBRA is progressing from a 3D-printed PLA prototype to a second prototype, planned to be made of titanium. The original prototype's configuration did not provide enough torque to support a titanium prototype.
- **Solution**: Acted as summer lead to organize undergraduate team of 5 into a rigging, coupling, and motors team to test multiple motor configurations to see what configuration would be best for prototype 2. Acted as part of the coupling team, where I researched COTS options for jaw coupler and elastomer and ultimately decided we needed custom parts. Designed coupler's face and estimated costs for CNC.
- **Results**: 3D printed coupler was used in weighted testing, where the face-to-face motor configuration achieved double the attainable torque.



Original configuration with COTS elastomer

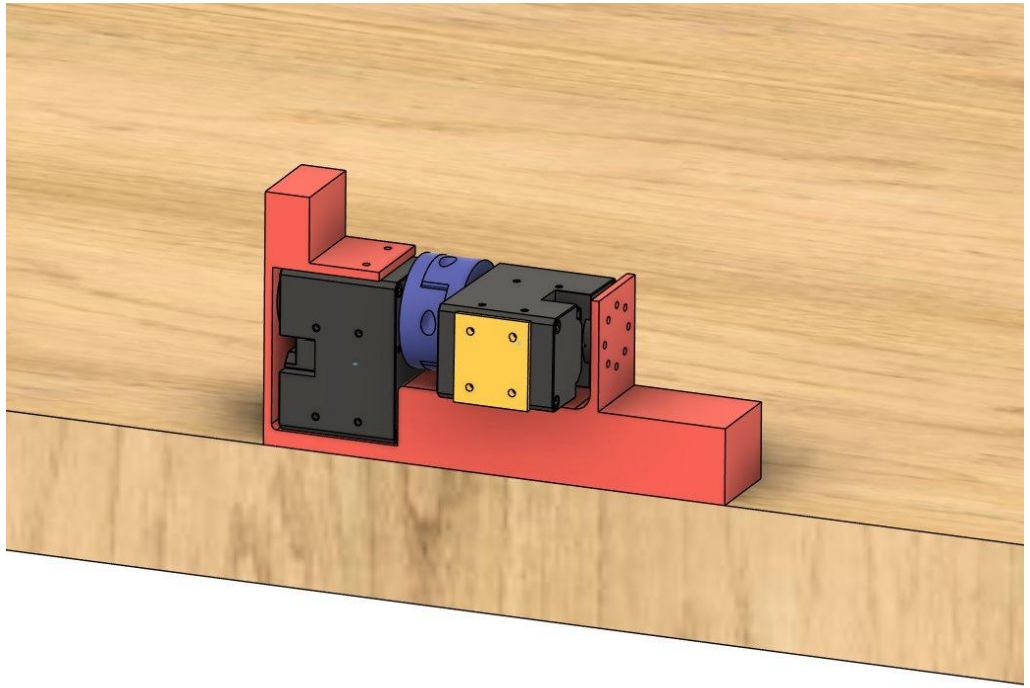


Jaw Coupler

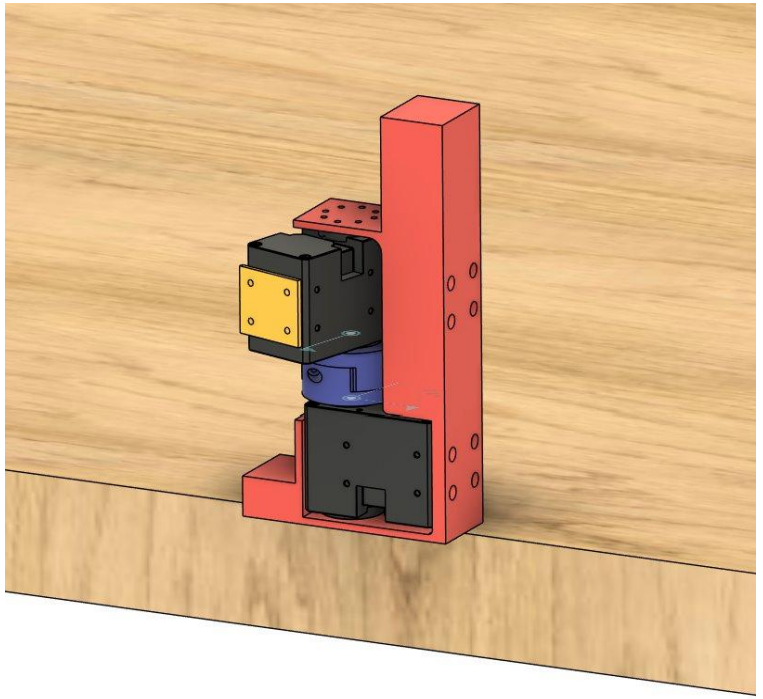


Final Configuration

COBRA Coupler Testing Configuration



Configuration 1: Torsion



Configuration 2: Axial

COBRA: Wire Routing

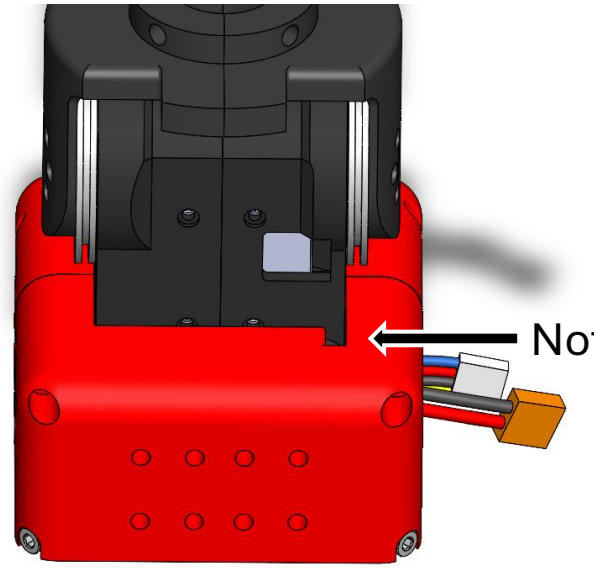
- **Problem**: The first prototype of COBRA was created without a wire-routing scheme, making assembly and maintenance of the robot difficult. As seen on the right, the power and data wire must traverse through the entirety of the robot.
- **Solution**: Created an updated wire routing scheme and used SolidWorks to add notches in body modules for the wires to pass through. Re-wired the robot.



COBRA: Wire Routing

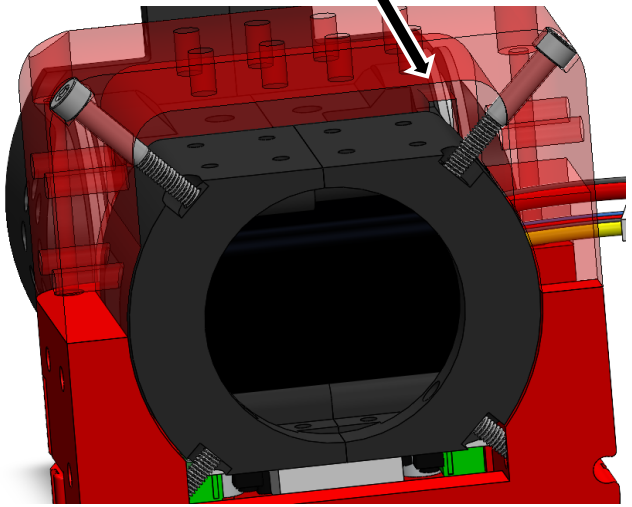


Notch for power wire



Notch for data wire

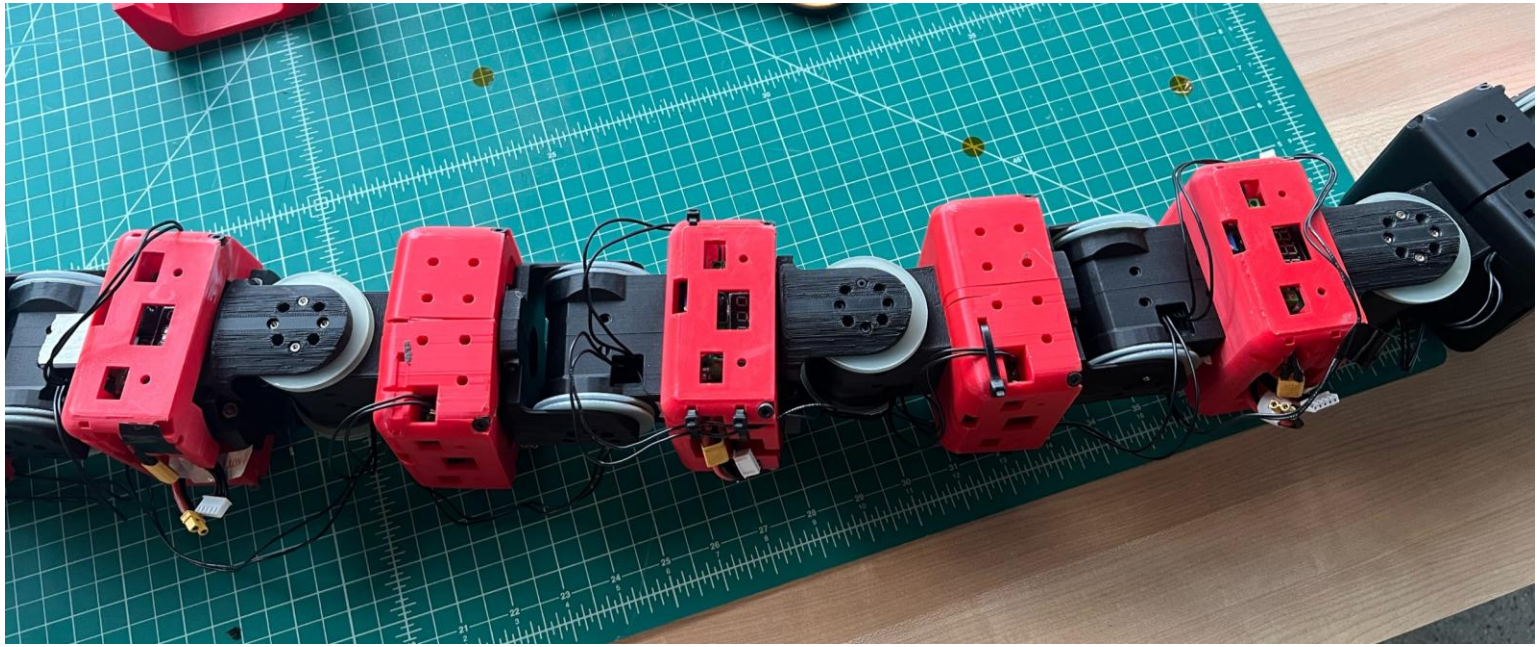
Bottom is open; allows for needed slack as body modules move



COBRA: Wire Routing



Before



After

COBRA Summit Fellowship

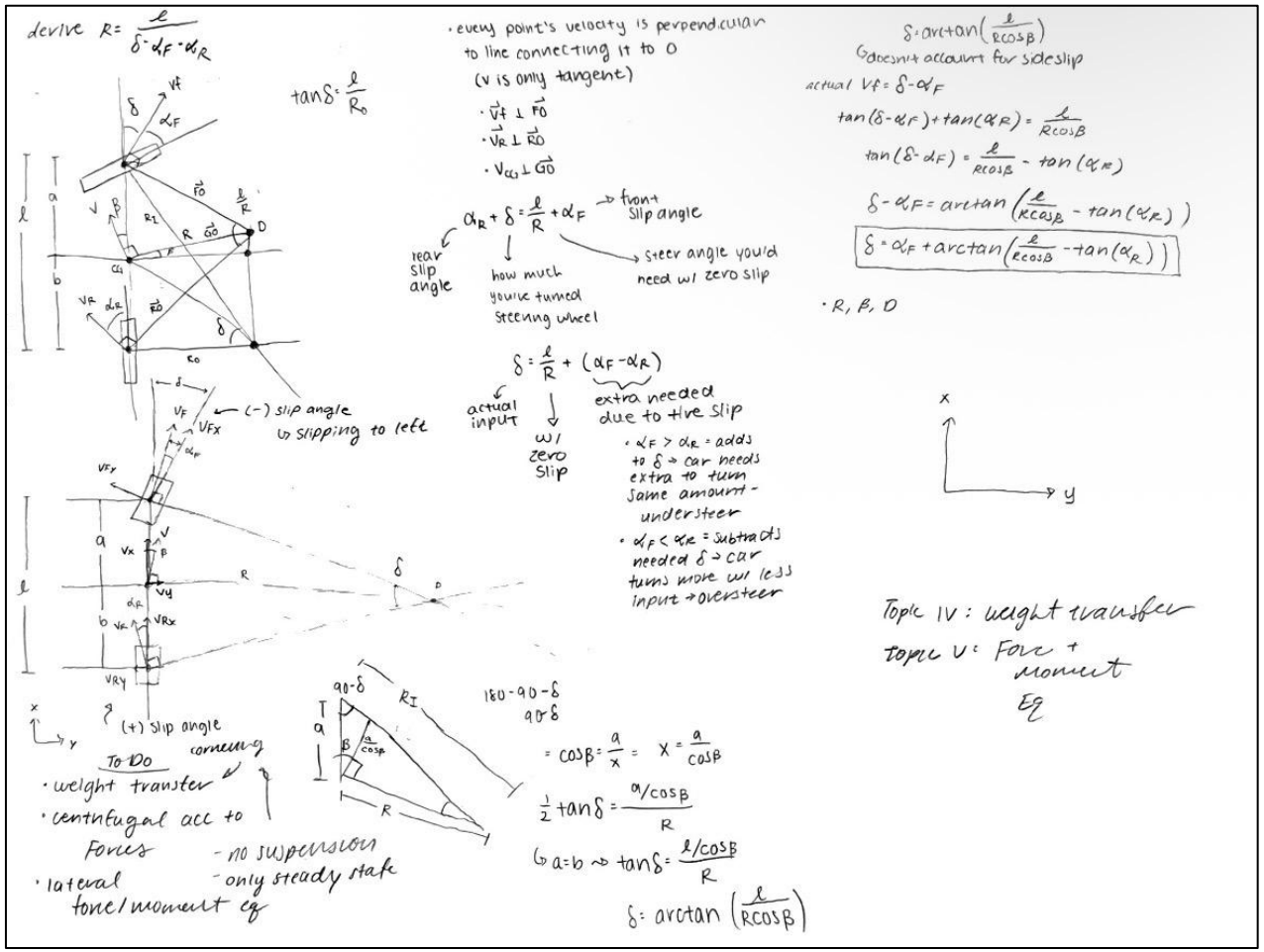
Project Proposal

- 2-page project proposal that won $2000 in funding for developing second prototype of COBRA
- Outlined goals and methods of designing the titanium 3D-printed prototype and the project's significance
- One of 80 students to win Northeastern University's PEAK Summit Fellowship
- Fall 2025 Project Proposal

Timetable

- 3-page timetable that outlined each week of Fall 2025 semester
- Determined what were the goals, steps, and expected time commitment of each week
- Fall 2025 Timetable

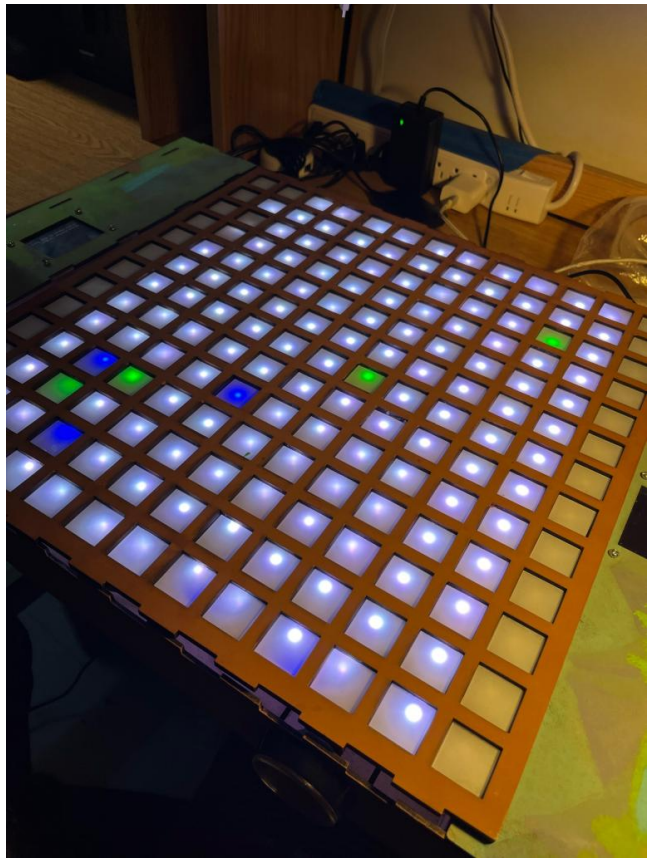
Vehicle Dynamics Modeling

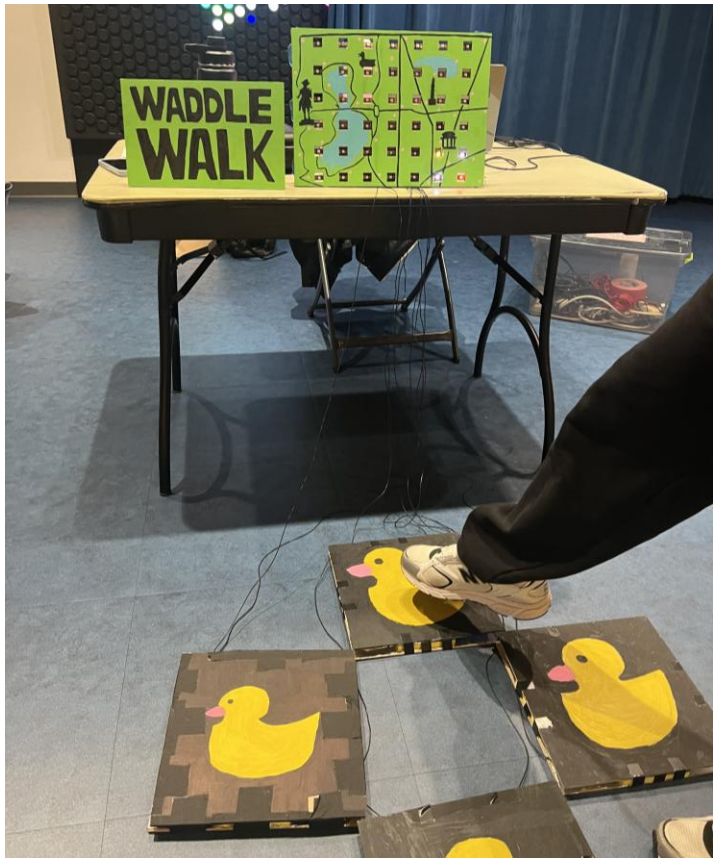


The goal of this research opportunity is to ultimately build a Python-based vehicle dynamics model. This position has just started, and I'm beginning with understanding the fundamentals and geometry of vehicles. My current task is to derive vehicle sideslip angle equations while accounting for independent front-wheel slip angles.

Projects

In Northeastern University's Engineering Course, I had 3 projects that allowed me to work in a group of engineers to make a solution. Given our problem statement, we brainstormed, designed, assembled, and presented our results. The first was a mini golf course, the second a portable board game, and the third was an educational game designed for children. This was in my first-year, and I was introduced to the reality of engineering for the first time. I learned vital lessons like how to use SolidWorks, how to design around a customer's needs, and how to keep factors like ease of assembly and durability in mind.





Waddle Walk

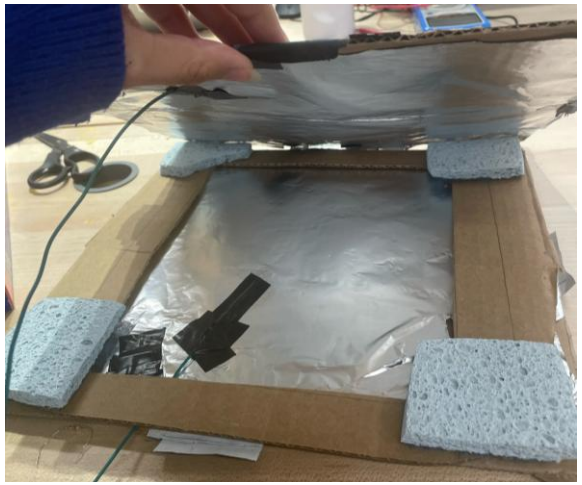
Goal: Our goal was to design, create, and run an educational children's game based on the Boston area for children at the Boston Children's Museum.

Process: Our game was centered around pressure plates that the kids would step on to "explore" an interactive LED map of the Boston Commons. We made pressure plates with laser cut 1/8” MDF and sponges to act as spacers. Aluminum foil was our conductive material, and when the kids stepped on the MDF and made the foil touch, the circuit was completed and the pressure plate was activated. Stepping on these pressure plates moved a light on an LED light strip on a map, effectively allowing the kids to move around the map via pressure plate.

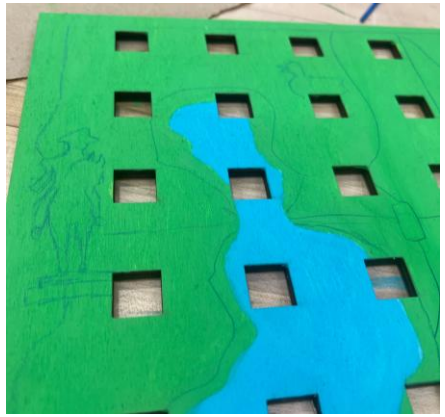
Results:
Our game was a huge success! One of our biggest concerns were the durability of our pressure plates, and we got through 3+ hours of children stomping on our plates with no spare parts used. Additionally, we appealed to an age group from 3 to 12 years old, which was also one of our considerations when designing the game. Finally, when we asked the kids how much they enjoyed the game out of 5, we averaged a score of 4.7, where we were aiming for at least a 4.5.

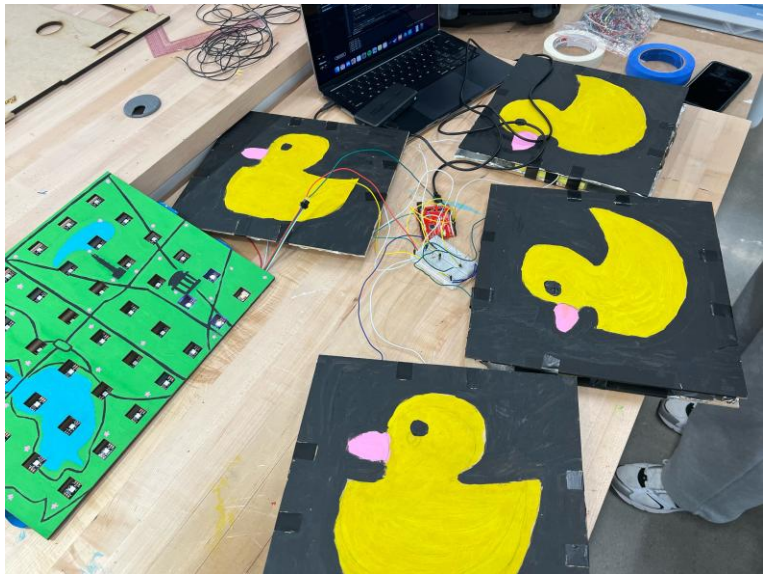


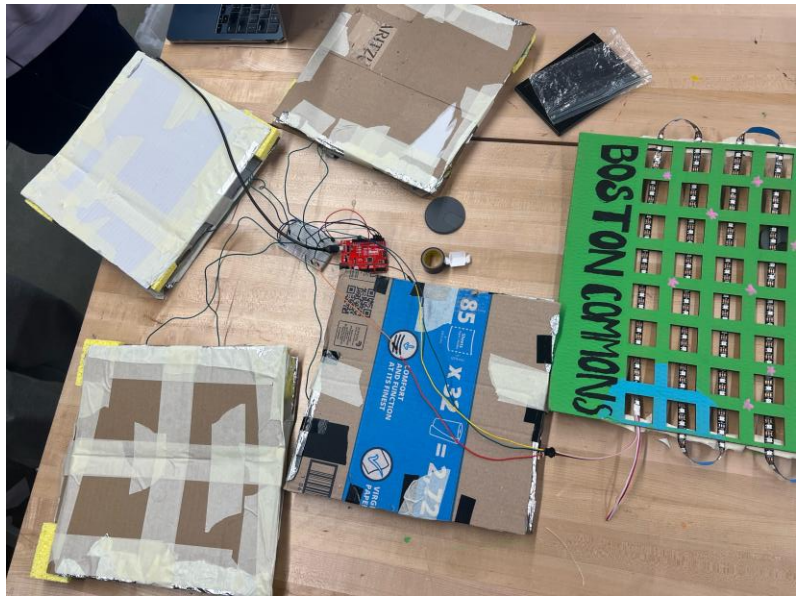
Waddle Walk: Prototype to Final Result



Here we can see the whole process; one major change we made was to the map. We altered the design so that less of the LED strip would show.







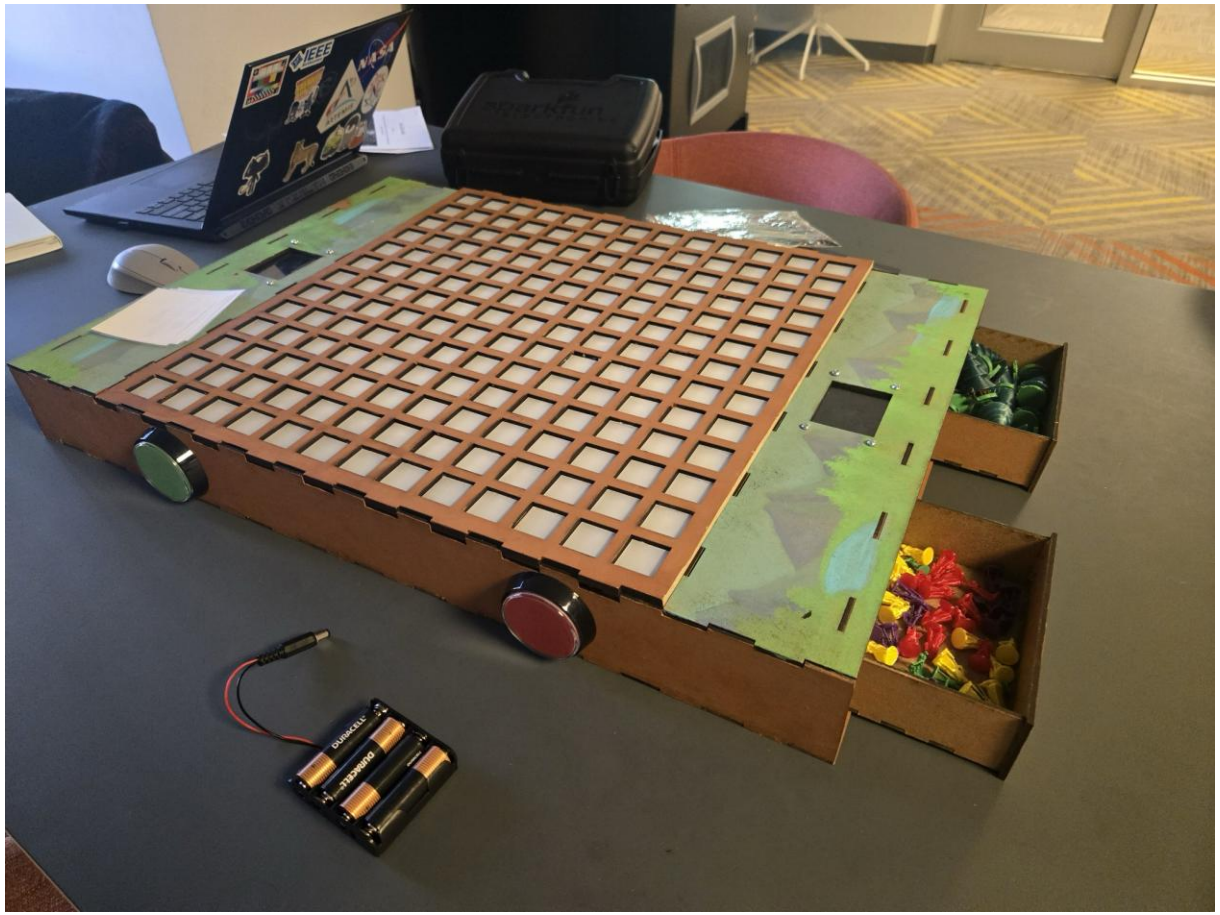




Race to the Summit

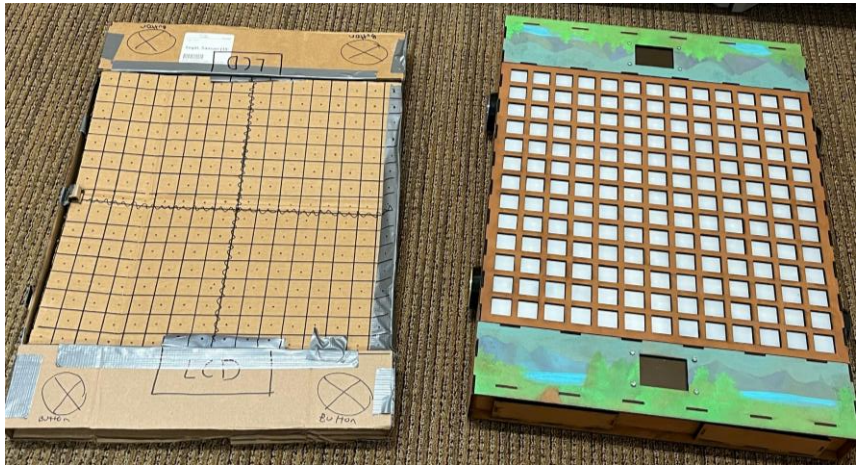
Goal: Our goal was to create a portable board game for a client.

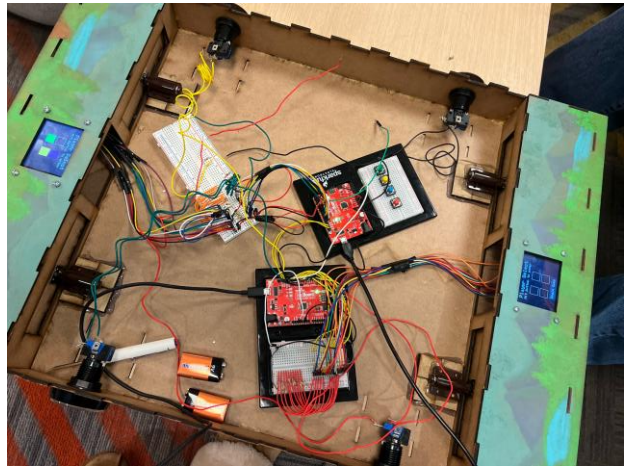
Process: We began with interviewing our client to find our constraints and to understand what she expected. After brainstorming, “Race to the Summit” was made, where 4 players push a button to get a randomly generated number, which is the number of spaces they can move. After every turn, green and blue spaces appear on the board. Green means that the space has grown a tree and a player cannot move there anymore, and if a player lands on a blue space, they receive a power up. The board is all laser-cut MDF with led lights shining through opaque acrylic. The game works with buttons, an LCD screen, and 3D-printed figures (player pieces and the trees that grow).

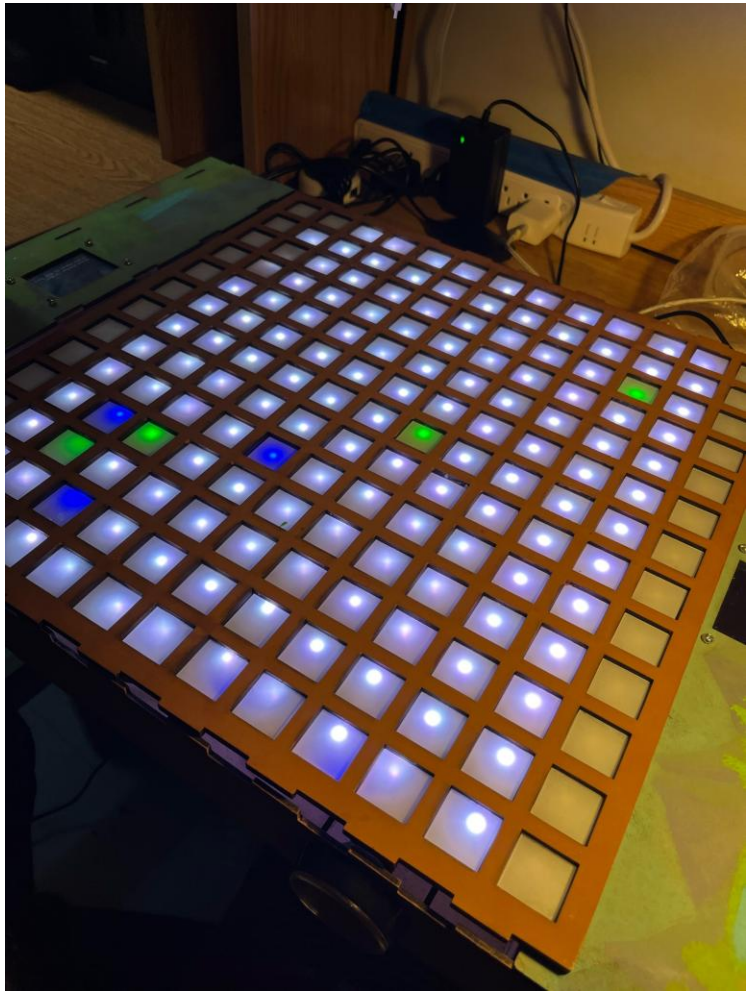


Results:
The client loved the game, and the game was also shown at Northeastern University’s College of Engineering Fair where many people stopped by to play the game.

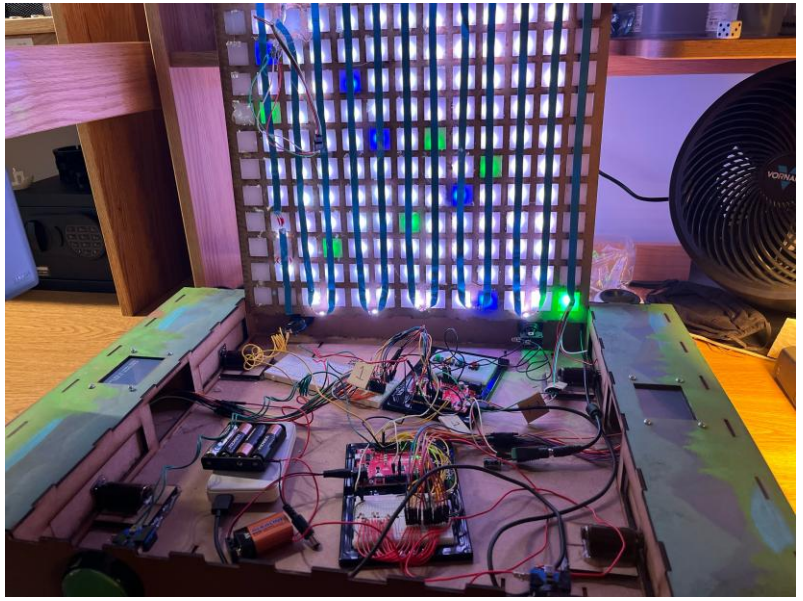
Race to the Summit: Prototype to Final Result



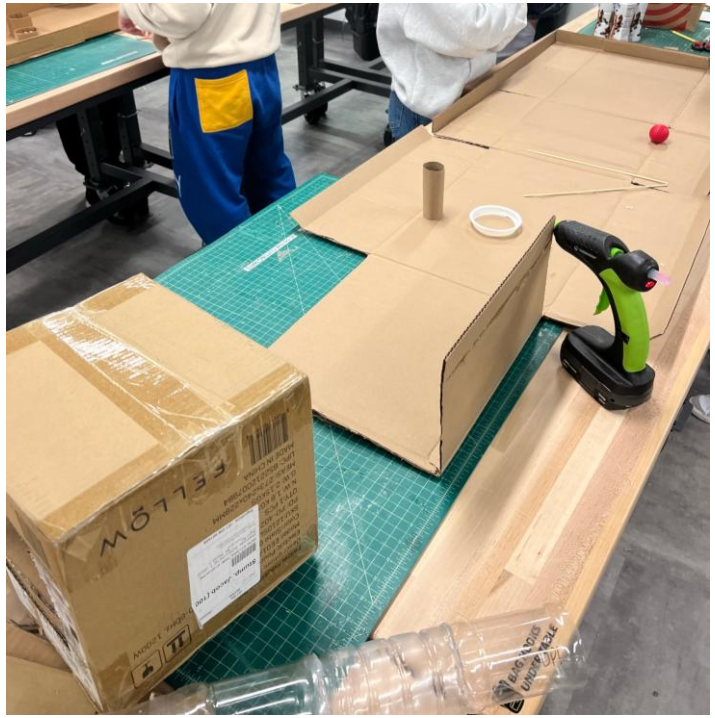




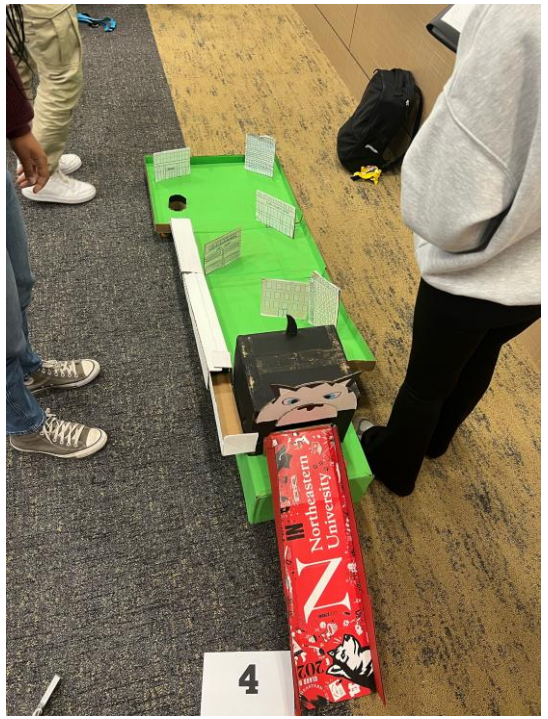


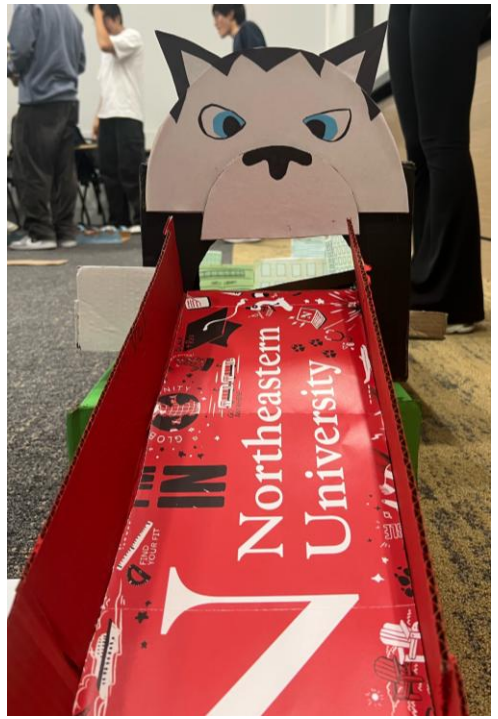


Mini Golf Course









Assembly → Trick Shot Ramp → Final Result!

Goal: Create a mini golf course to be played by the class.
Process: After the initial brainstorming stage, we collected our materials (cardboard) and assembled.
Result: Our mini golf course worked as designed, and we even had a player successfully make the trick shot! This was achieved by hitting the ball at the right velocity where it would slide down the initial ramp rather than go through the rest of the course in green.

Contact

- Phone Number: 602-354-0424
- Email: Arimori.s@northeastern.edu
- LinkedIn: https://www.linkedin.com/in/sera-arimori

Links

- COBRA's background: COBRA's first prototype wins NASA's BIG Idea Challenge
  - Describes what COBRA has accomplished in the past, and what system our current team is improving upon.
- Fellowship Awardees: Fall 2025 PEAK Experiences Awardees
- Published article: Utilizing voltage to slow coastal erosion – NU Sci Magazine
  - An article that introduces the idea of using low voltages to fortify our eroding shores.